

HARSHIT GOYAL

VLSI & Semiconductor Engineer — B.Tech Electrical Engineering, IIT Delhi

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EDUCATION

Indian Institute of Technology Delhi <i>B.Tech, Electrical Engineering; CGPA: 7.513</i>	2022 – 2026
Bharat Mata Saraswati Bal Mandir Sr. Sec. School <i>Class XII (CBSE): 94.8%</i>	2022
K.M. Public School <i>Class X (CBSE): 95.0%</i>	2020

EXPERIENCE

AI Engineer Intern <i>Samsung R&D Institute</i>	Jul 2026 – Present Delhi
Training and fine-tuning Vision-Language Models (VLMs), building custom solutions targeted at market-driven product requirements.	
ASIC Synthesis Intern <i>Cadence Design Systems</i>	May 2025 – Jul 2025 Noida
- Drove Verilog RTL-to-Pre-CTS synthesis flows for multiple design blocks in Cadence Genus and Innovus, taking designs from elaboration and logic synthesis through floorplanning and placement. - Analyzed setup/hold timing slack, area, power, and standard-cell (library-cell) utilization at each stage, comparing QoR across runs to isolate bottlenecks and drive constraint and optimization tuning. - Built a Python netlist-parsing tool to extract per-instance cell-utilization and QoR metrics from mapped netlists, replacing manual inspection of synthesis reports. - Migrated legacy TCL synthesis-automation scripts to Python, cutting execution time and improving maintainability and reuse across flows.	
Research Intern <i>Cape Peninsula University of Technology</i>	Dec 2023 – Jan 2024 Cape Town
Researched AI-enabled smart-microgrid architectures and embedded/edge deployment constraints, building a structured evaluation framework and authoring a 32-source technical report to guide adoption.	
Technology Consulting Intern <i>PwC India</i>	Jun 2024 – Jul 2024 Gurugram
Digitized financial-service workflows for a PACS cooperative network serving 130M+ farmers and automated the underlying data processing in Python, cutting end-to-end processing time by 15%.	

PROJECTS

Codebase Intelligence Platform – Graph-Based Static Analysis <i>tree-sitter · ArcadeDB · ChromaDB · FastAPI</i>	github.com/goyal-harshit
- Ingests any Git repository, parsing source with tree-sitter into a language-agnostic dependency graph (ArcadeDB) plus a semantic vector index (ChromaDB) for structural and natural-language queries. - Detects code-quality issues including god objects, dead code, and circular dependencies by traversing call and import relationships in the graph. - Parsed a 24-file Flask codebase in 0.28s, extracting 388 functions and 53 classes; exposed the analysis through a FastAPI service.	
QuantAI – Quantitative Investment Research Platform <i>FastAPI · Next.js · PostgreSQL/TimescaleDB · Docker</i>	
- Built a full-stack quant research platform for NSE/BSE equities with a multi-factor screener and an event-driven backtester over TimescaleDB time-series storage. - Added a portfolio tracker and a RAG-grounded AI assistant with a swappable LLM backend to answer research queries over market data. - Containerized the stack with Docker and set up GitHub Actions CI/CD for automated build, test, and deployment.	
AI/ML Fantasy Cricket Predictor – 9th Nationally, FIFS Gameathon 2025	Google Cloud & Dream11
- Built an automated CricSheet data pipeline with match-wise feature engineering to produce per-player predictive features from raw ball-by-ball data. - Trained an ensemble of Random Forest, Gradient Boosting, and a PyTorch player-embedding network to model player performance, placing 9th nationally.	

TECHNICAL SKILLS & ACHIEVEMENTS

Digital & VLSI: Verilog, RTL Design, Logic Synthesis, Static Timing Analysis (STA), CMOS, CPLD/FPGA

EDA & Tools: Cadence Genus, Cadence Innovus, LTSpice, GVim, Git/GitHub Actions, Docker

Programming: Python, TCL, C++, MATLAB, Verilog

ML & Software: PyTorch, TensorFlow, FastAPI, scikit-learn

Achievements: JEE Advanced 2022 – AIR 530 (GE) of 250,000+; JEE Main 2022 – 99.74 percentile of 1,000,000+